

Homework 6: Making Your Own ggplot

GSND 5345Q, Fundamentals of Data Science

Multiple Due Dates (Video: Thu 2/26; Comments/write-up: Monday 3/2)

Objective

Demonstrate your understanding of data visualization principles by creating an informative and well-annotated plot using `ggplot` in R. You will present your plot in a recorded video and engage with your peers by providing constructive feedback on their visualizations.

Instructions

1. Find a Dataset

Choose one of the following:

- A dataset provided in class.
- A dataset from your own research or work experience.
- A publicly available dataset from a credible source (Kaggle, government databases, academic repositories).

2. Create an R Markdown Report (50 points)

- Use `ggplot2` to create **at least one** well-designed and well-annotated plot (or multiple plots) that meaningfully describe your dataset.
- Apply best practices in data visualization, such as:
 - Choosing an appropriate plot type for your data.
 - Using clear labels, titles, and legends.
 - Selecting effective color schemes.
 - Showing the data in a way that avoids misleading interpretations.
 - Encoding three or more variables when possible.
- Provide a short written explanation of your plot, addressing:
 - What the plot shows and why it's appropriate for your data.
 - Which visualization principles were applied.
 - Any design choices made to improve readability or clarity.
- Include your **fully reproducible code** in an R Markdown (`.Rmd`) file.

3. Video Presentation (50 points)

- Record a **3-5-minute video** explaining your visualization.
- Your video should include:
 - A brief introduction to your dataset.
 - A walkthrough of your plot(s), describing the insights they reveal.
 - An overview of the code used to generate the plot.
 - A discussion of the **data visualization principles applied** in your work.
- Post your video in the Chat section on Canvas no later than **11:59 PM on Thu, 2/26**.

4. Peer Review (50 points)

- Watch **at least five** of your classmates' videos.
- Provide **thoughtful and constructive comments** on each, focusing on:
 - Strengths of their plot design and presentation.
 - Suggestions for improvement (if applicable).
- Document your comments in your `.Rmd` file (include the names of the videos you reviewed).
- Submit the updated `.Rmd` file with your own plot code/comments and your peer review comments on Canvas no later than **11:59 PM on Monday, 3/2**.

Grading Breakdown

Category	Points
Well-designed ggplot with clear annotations and applied principles	50
Video presentation with clear explanation and visualization discussion	50
Peer review with meaningful feedback on five classmates' videos	50
Total	150

Additional Notes

- Ensure your plots and video are accessible and understandable to someone unfamiliar with your dataset.
- Be concise but thorough in your explanations.
- Use proper formatting and grammar in your `.Rmd` file and comments.
- Engage respectfully and constructively in your peer reviews.

Happy plotting!